

EXPERIMENT 1 a:

You have received a suspicious Windows executable file named suspicious.exe, Without running it, identify unique **patterns** or **signatures** that could indicate its **malware family** or type. Use an open-source tool to create rules or signatures that can help classify this file among others. A basic generic **YARA** rule has already been created to detect common Windows malware indicators such as process injection APIs and command execution strings.

Tool: YARA

Important Disclaimer: We will NOT use live malware binaries. That would be unsafe and unethical. Instead, we use malware *signatures* embedded into test binaries.

The main goal is to

- ✓ Identify and classify a suspicious binary file WITHOUT executing it, by using static patterns (strings, headers, APIs) and YARA rules.
- ✓ NOT infecting the system
- ✓ NOT running malware
- ✓ Only matching patterns

YARA is an open-source tool used to identify and classify malware based on textual patterns, strings, and binary characteristics.

By performing static analysis, malware behavior can be inferred without running the file, ensuring system safety.

What are we Analyzing:

We analyze binary files (mostly .exe) and look for:

- Malware-specific strings
- Windows API function names
- PE(Portable Executable) file headers
- Registry or command indicators

These indicators help us guess the malware type.

Why Yara Tool??

YARA works like pattern matching:

“If this file contains these strings / structures,
then classify it as this type of malware.”

Procedure (Steps for performing experiment)

In Kali Linux VM

✓ Update and Upgrade

```
sudo apt update && sudo apt upgrade -y
```

✓ Install yara

```
sudo apt install yara -y
```

✓ yara version

```
yara --version
```

1. Use a known test malware sample:

Creating Sample file

✓ Go to kali linux browser and search for EICAR.com

✓ In eicar website.. chose eicar.com and download the file.

✓ Make sure the contents of downloaded file paste in below exe file(sample.exe)

```
vi sample.exe
```

```
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
```

```
Save and Exit
```

Or

Create a new exe file

```
echo "Student malware lab test: EICAR sample" > simple.exe
```

2. Then create yara rules:

Basic Yara Rules

```
vi eicar_rule.yar
```

```
rule EICAR_Test_File
{
  meta:
    author = "Lab Student"
    description = "Detects EICAR antivirus test file"
    reference = "Standard test signature"
  strings:
    $eicar_full = "EICAR-STANDARD-ANTIVIRUS-TEST-FILE" ascii wide nocase
    $eicar_part = "EICAR" ascii wide nocase
  condition:
    any of ($eicar*)
}
```

3. Run:

```
yara eicar_rule.yar sample.exe
```

Then expected output looks like below

```
EICAR_Test_File eicar_sample.exe
```

Possible Results

- No output → benign PE (expected for nc.exe)
- Rule name printed → suspicious indicators found

Both results are valid for the experiment.

Result

- The suspicious binary contains known malware-related strings

- YARA rules successfully detected the file based on static signatures
- No execution of the file was required.